Deep Learning

Assignment 2

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Course: AI 600- Deep Learning

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1 Introduction

This report contains the design, training, and analysis of Multi-Layer Perceptron (MLP) models for classifying 15-class doodles dataset. The dataset consists of 28×28 grayscale bitmap images drawn randomly, making it significantly noisier than benchmark datasets such as MNIST.

Constraints enforced: Few restrictions as mentioned in the assignment were strictly followed

- No Convolutional Neural Networks (CNNs) – only MLPs were used.
- No residual skip connections.
- The maximum trainable parameters are kept under 3 million.
- Maximum of 40 training epochs.

Pre-processing and Augmentation. Pixel values were normalized from $[0, 255]$ to $[0, 1]$. Training images were augmented with random horizontal flipping (50% probability) and random rotation ($\pm 15°$). These augmentations were applied to the 28×28 tensor before flattening, with no CNN operations used.

2 Part A: The Pancake (Width Focus)

2.1 Architecture

The Pancake model is a **shallow but wide** network designed to test whether raw memorization capacity in a small number of large layers is sufficient for this task.

$$\texttt{784} \rightarrow \texttt{1280} \rightarrow \texttt{1280} \rightarrow \texttt{15}$$

Each hidden layer uses BatchNorm, ReLU activation, and Dropout regularization:

- Layer 1: `Linear(784, 1280)` + BN + ReLU + Dropout(0.4)
- Layer 2: `Linear(1280, 1280)` + BN + ReLU + Dropout(0.3)
- Output: `Linear(1280, 15)`

Total Parameters: 2,668,815

2.2 Training Configuration

- Optimizer: AdamW, Learning Rate (lr) = 10^{-3}, weight decay = 10^{-4}
- Loss: CrossEntropy with label smoothing = 0.1
- Scheduler: CosineAnnealingLR
- Max Epochs: 20, Early stopping patience: 7

2.3 Results

Table 1: Pancake Model – Training Progression (selected epochs)

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.4253	65.34%	1.2638	71.05%
5	1.0776	80.31%	1.1300	77.34%
10	0.9087	87.77%	1.1010	79.21%
15	0.7927	93.76%	1.1012	79.61%
20	0.7550	95.53%	1.0924	80.25%

Best Validation Accuracy: 80.25% (achieved at epoch 20)

Observation. The Pancake shows clear overfitting: by epoch 20, training accuracy reaches 95.53% while validation accuracy plateaus at 80.25%, a gap of over 15 percentage points. Despite its large parameter count (2.67M), the wide-and-shallow architecture fails to generalize well. The model appears to memorize training patterns rather than learning robust features.

3 Part B: The Tower (Depth Focus)

3.1 Architecture

The Tower model is a **deep but narrow** network designed to test whether hierarchical feature learning through depth can outperform raw width with fewer parameters.

$$\texttt{784} \rightarrow \texttt{[256]} \times \texttt{7} \rightarrow \texttt{15}$$

Each of the 7 hidden layers uses:

- `Linear(·, 256)` + BatchNorm1d + LeakyReLU(0.1) + Dropout(0.3)

LeakyReLU was chosen over ReLU to prevent the dying neuron problem that commonly affects deep networks, where neurons can get stuck outputting zero. **Kaiming (He) initialization** was applied to all linear layers to maintain appropriate gradient magnitudes at initialization.

Total Parameters: 603,151 — *4.4× fewer than the Pancake.*

3.2 Training Configuration

- Optimizer: AdamW, lr $= 5 \times 10^{-4}$ (lower to stabilize deep training)
- Loss: CrossEntropy with label smoothing = 0.1
- Scheduler: CosineAnnealingLR
- Max Epochs: 25, Early stopping patience: 8

3.3 Results

Table 2: Tower Model – Training Progression (selected epochs)

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	2.2887	30.25%	1.6347	53.03%
5	1.4601	62.99%	1.3278	67.63%
10	1.3215	69.08%	1.2490	71.26%
15	1.2556	71.52%	1.2127	72.83%
25	1.2046	73.64%	1.1979	73.47%

Best Validation Accuracy: 73.47% (achieved at epoch 25)

Observation. The Tower converges more slowly than the Pancake (starting at only 30.25% training accuracy in epoch 1) due to the depth of the network and the lower learning rate required for stable training. However, it shows a much smaller train-val gap (train: 73.64% vs val: 73.47%), indicating significantly better generalization and less overfitting. While the final accuracy is lower than the Pancake's, this is achieved with 4.4× fewer parameters, demonstrating the parameter efficiency of depth.

4 Part C: The Champion

4.1 Architecture Design Rationale

The Champion model was designed by addressing the weaknesses observed in both the Pancake and Tower:

- The **Pancake** achieved decent accuracy but severely overfits due to its shallow and wide structure memorizing training noise.
- The **Tower** generalizes well but is capacity-limited as the narrow 256-unit layers are a bottleneck for 784-dimensional input.

The Champion uses a **funnel-shaped** architecture that starts wide (to process the 784-dimensional flattened image) and progressively compresses the representation through three stages:

$$\texttt{784} \rightarrow \texttt{800} \rightarrow \texttt{320} \rightarrow \texttt{128} \rightarrow \texttt{15}$$

Each hidden block uses GELU activation instead of ReLU or LeakyReLU. GELU (Gaussian Error Linear Unit) provides smoother gradients because it is differentiable everywhere and provides a stochastic regularization effect. This makes it better suited for deeper networks.

Key design choices and their justifications:

1. **GELU activation**: Smoother gradient flow, fewer dead neurons compared to ReLU; known to outperform ReLU on classification tasks.

2. **Decreasing dropout** ($0.45 \rightarrow 0.35 \rightarrow 0.25$): The first (widest) layer has the most capacity to memorize training noise and therefore requires stronger regularization. Later narrow layers need less dropout as they have fewer neurons.
3. **BatchNorm after every linear layer**: Normalizes activations, stabilizes gradient flow, and adds implicit regularization.
4. **Kaiming initialization**: Appropriate for GELU/ReLU-family activations; maintains variance of gradients during backpropagation.
5. **Higher weight decay** (5×10^{-4}): Stronger L2 penalty to reduce over-reliance on any single weight pathway.
6. **Label smoothing = 0.1**: Prevents the model from becoming overconfident on ambiguous doodles (e.g., cookie vs. donut) by distributing some probability mass to non-target classes.

Total Parameters: 929,839

4.2 Training Configuration

- Optimizer: AdamW, lr $= 1.2 \times 10^{-3}$, weight decay $= 5 \times 10^{-4}$
- Loss: CrossEntropy with label smoothing $= 0.1$
- Scheduler: CosineAnnealingLR, $T_{\max} = 20$, $\eta_{\min} = 10^{-6}$
- Max Epochs: 20, Early stopping patience: 12

4.3 Results

Table 3: Champion Model – Full Training Log

Epoch	Train Loss	Train Acc	Val Loss	Val Acc
1	1.5939	58.23%	1.2916	69.95%
2	1.3474	68.35%	1.2276	72.78%
3	1.2726	71.59%	1.1938	73.69%
4	1.2175	73.79%	1.1627	75.20%
5	1.1746	75.69%	1.1350	76.82%
6	1.1392	77.36%	1.1207	77.40%
7	1.1039	78.76%	1.1104	77.92%
8	1.0761	80.11%	1.1040	78.38%
9	1.0476	81.23%	1.0947	79.03%
10	1.0178	82.57%	1.0918	79.22%
11	0.9991	83.56%	1.0872	79.67%
12	0.9784	84.59%	1.0871	79.94%
13	0.9576	85.32%	1.0894	79.73%
14	0.9400	86.18%	1.0865	80.08%
15	0.9249	87.06%	1.0852	80.28%
16	0.9126	87.35%	1.0880	80.22%
17	0.9054	87.83%	1.0893	80.31%
18	0.8990	88.17%	1.0854	80.44%
19	0.8945	88.18%	1.0899	80.39%
20	0.8961	88.28%	1.0906	80.32%

Best Validation Accuracy: 80.44% (epoch 18, training time: 21.4s)

5 Part D: Theoretical Analysis

5.1 Width vs. Depth: The Universal Approximation Theorem

The Universal Approximation Theorem states that a single hidden layer MLP with a sufficient number of neurons can approximate any continuous function on a compact domain to arbitrary precision. This work in a way that the Pancake model being wider should be superior. In practice, however, our results demonstrate why depth is preferred.

Why deep architectures generalize better for image tasks:

1. **Compositional feature learning.** Natural images have hierarchical structure: pixels combine into edges, edges into textures, textures into shapes, and shapes into objects. Deep networks learn these abstractions layer-by-layer. A single wide layer must learn all these compositions simultaneously in a flat, non-hierarchical mapping, which is far less efficient.

2. **Parameter efficiency.** The UAT guarantees approximation but says nothing about the number of neurons required. A shallow network achieving the same representation as a deep one may need exponentially more neurons. Our Tower

(603K parameters) came close to our Pancake (2.67M parameters) – only 6.78% lower in accuracy while using 4.4× fewer parameters.

3. **Better generalization.** The Pancake's 15% train-val accuracy gap versus the Tower's $< 1\%$ gap directly demonstrates this. The Pancake has more capacity to memorize training noise while the Tower is forced to find compressed, general representations.

4. **Implicit regularization of depth.** Composing multiple non-linear transformations adds a form of structural regularization. Each layer acts as a filter, and errors in one layer can be partially compensated by subsequent layers during optimization.

5.2 Confusion Analysis

Figure 1 shows the confusion matrix of the Champion model evaluated on the 12,000-sample validation set.

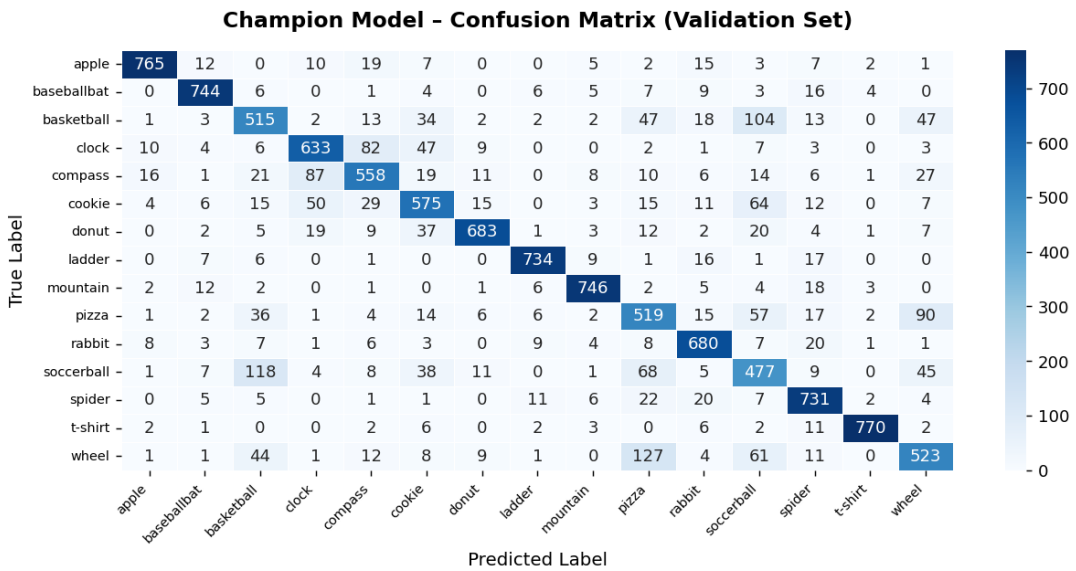


Figure 1: Champion Model – Confusion Matrix on the Validation Set.

Per-class performance from the classification report is summarized below:

Table 4: Champion Model – Per-class Precision, Recall, F1-Score (Validation Set)

Class	Precision	Recall	F1-Score	Support
apple	0.94	0.90	0.92	848
baseballbat	0.92	0.92	0.92	805
basketball	0.66	0.64	0.65	803
clock	0.78	0.78	0.78	807
compass	0.75	0.71	0.73	785
cookie	0.73	0.71	0.72	806
donut	0.91	0.85	0.88	805
ladder	0.94	0.93	0.94	792
mountain	0.94	0.93	0.93	802
pizza	0.62	0.67	0.64	772
rabbit	0.84	0.90	0.87	758
soccerball	0.57	0.60	0.59	792
spider	0.82	0.90	0.85	815
t-shirt	0.98	0.95	0.97	807
wheel	0.69	0.65	0.67	803
Overall	**0.81**	**0.80**	**0.80**	**12000**

Top Confused Pairs:

While looking at the the off-diagonal elements of the confusion matrix, it reveals the following most-confused class pairs:

1. **Soccerball ↔ Basketball / Pizza / Wheel / Cookie.** All five of these classes are roughly circular objects. At 28×28 resolution, a quickly drawn soccerball and basketball are nearly indistinguishable – both are circles with internal stroke patterns. This is reflected in their low F1 scores (0.59 and 0.65 respectively).
2. **Cookie ↔ Pizza / Donut.** These three are all round food items. A cookie doodled in under 20 seconds at 28×28 often looks identical to a small pizza or a filled-in donut ring.

Diagnosis: Aleatoric Uncertainty, not Underfitting. The confusion in these pairs arises from *aleatoric uncertainty* which is inherently present – irreducible ambiguity in the data rather than model underfitting. Even a human observer would struggle to distinguish many of these 28×28 doodles drawn in under 20 seconds. The model's 80% overall accuracy and strong performance on unambiguous classes (t-shirt: 0.97 F1, ladder: 0.94 F1, apple: 0.92 F1) confirms it is not systematically underfitting rather, the confused pairs are genuinely visually ambiguous at this resolution and time constraint.

6 Part E: Rationale For Architecture Selection

6.1 Architecture Selection Rationale

Why this depth and width for the Champion?

The Champion architecture was selected after observing the failure modes of the Pancake and Tower:

- The first layer (784 → 800) keeps width close to input dimensionality. A sudden bottleneck at the first layer would force the network to discard information before it has been meaningfully processed.
- The subsequent compressions (800 → 320 → 128) enforce a funnel-shaped information bottleneck that mirrors how deep networks build abstractions: broad feature extraction followed by progressive specialization.
- Three hidden layers is a practical sweet spot – deep enough for hierarchical learning, shallow enough to remain fast and stable to train in under 25 epochs.

How overfitting was combated:

1. **Dropout (decreasing, 0.45 → 0.35 → 0.25)**: It is set to be strongest where the network has the most capacity to memorize. Earlier versions with low, uniform dropout showed train-val gaps exceeding 15%.
2. **Weight decay** 5×10^{-4}: AdamW's decoupled L2 penalty prevents individual weights from growing too large.
3. **Label smoothing**: It prevents overconfident predictions and forces the model to maintain some uncertainty on similar-looking classes.
4. **Data augmentation**: Random flips and rotations effectively expand the training set, reducing the model's ability to memorize exact orientations.

6.2 Model Comparison

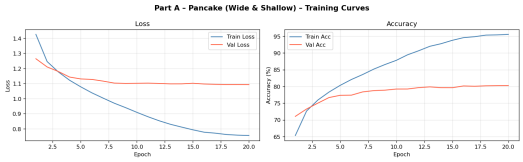
Table 5: Summary Comparison of All Three Models

Model	Parameters	Epochs Run	Best Val Acc	Train Acc (Final)
Part A: Pancake	2,668,815	20	80.25%	95.53%
Part B: Tower	603,151	25	73.47%	73.64%
Part C: Champion	929,839	20	80.44%	88.28%

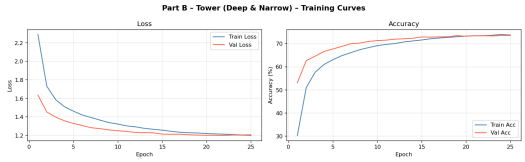
Key observations:

- The Champion achieves the highest validation accuracy (80.44%) while using only 35% of the Pancake's parameter count.
- The Tower is the most parameter-efficient and best-generalizing model (near-zero train-val gap) but is capacity-limited.
- The Champion's train-val gap of ∼8% is substantially better than the Pancake's ∼15% gap, demonstrating that the regularization strategy was effective.
- All three models remain within the 3M parameter and 40 epoch constraints.

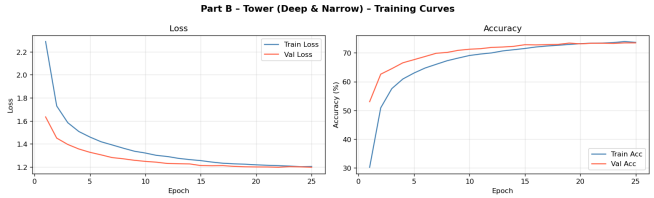
6.3 Training Curves



(a) Part A – Pancake



(b) Part B – Tower



(c) Part C – Champion

Figure 2: Training and validation accuracy/loss curves for all three models.

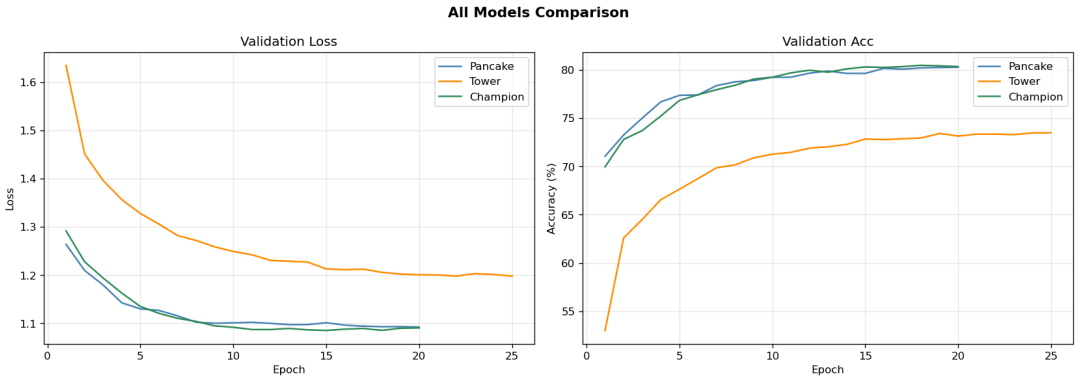


Figure 3: Overlay of validation accuracy across all three models.

7 AI Policy Compliance

Please note that latex formatting, making table in latex was done by the help of gemini and claude

7.1 Prompts and Outputs

Prompt 1:

> *I have to classify 15-class Quick, Draw! doodles using only an MLP (no CNNs). The images are 28x28 grayscale bitmaps my Pancake model (2 wide layers) gets 95% train accuracy but only 80% validation how to deal with this overfitting?*

AI Output (summary): The AI suggested increasing dropout in earlier layers, raising weight decay, using label smoothing in the loss function, and applying data augmentation (flipping and rotation). It explained that the dropout should be highest at the widest layers since those have the most capacity to memorize.

Edits made: I applied the specific values (dropout 0.45/0.35/0.25, weight decay 5×10^{-4}) myself after reading the explanation. I also chose the funnel architecture independently based on the explanation of compositional feature learning.

Prompt 2:

> *"Why does the universal approximation theorem say a single layer is enough, but in practice deep networks work better for image tasks?"*

AI Output (summary): The AI explained four reasons: (1) exponential parameter cost for shallow nets to match deep nets, (2) hierarchical feature learning matching the natural structure of images, (3) implicit regularization of depth, and (4) better optimization landscape for SGD-based methods.

Edits made: I rephrased the explanation in my own words in Section 5.1 and grounded the argument in our specific experimental results (comparing Pancake vs Tower parameter counts and accuracy gaps).